

Multilanguage Semantic Classification Using a Language Router

**KAZAKH-RUSSIAN SENTIMENT CLASSIFICATION
WITH CLASSICAL ML AND FASTTEXT BASELINES**

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Tools: Python,
scikit-learn,
fastText,
Kaggle/Jupyter



Problem Statement

Sentiment analysis for multilingual data is difficult because Kazakh and Russian texts differ in:

- alphabet and morphology
- word formation
- spelling variation
- informal review language
- code-switching between languages
- limited high-quality NLP resources for Kazakh

Main challenge:

Businesses need an automatic way to classify reviews as positive or negative.

Why it matters:

- Track customer satisfaction
- Detect problematic locations or services
- Analyze sentiment trends over time
- Support decision-making for local businesses

The project focuses on the Kazakhstan context, where multilingual Cyrillic text and code-switching make sentiment classification more challenging.

Project Goal:

Goal:

Build a binary sentiment classification system for Kazakh and Russian service reviews.

Classification task:

- Positive review
- Negative review

Main approach:

Compare classical machine learning models with transformer-based models and introduce a language-routing strategy for better multilingual performance.

Modeling stages:

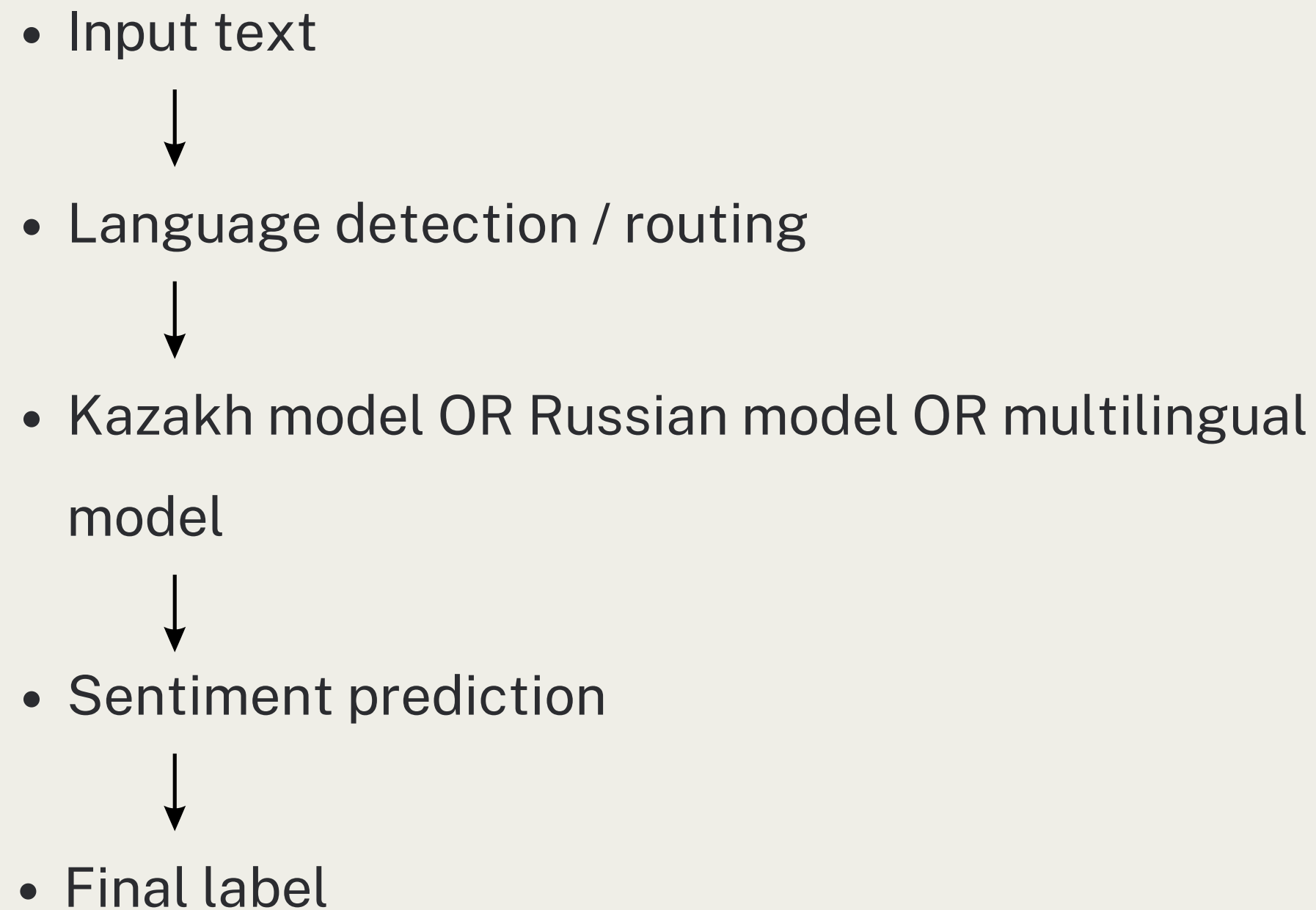
Train baseline TF-IDF models

Train transformer models

Apply language routing

Test on Kazakhstan-specific reviews

PROJECT IDEA: LANGUAGE ROUTER



What is a language router?

A language router is a component that detects or identifies the language of an input text and sends it to the most suitable model.

Why it is useful

- 1.improves specialization for each language
- 2.allows separate preprocessing strategies
- 3.reduces noise from mixed-language training
- 4.makes the system easier to extend to more languages

DATASETS USED

Dataset	Language	Purpose
KazSAnDRA	Kazakh	Training and evaluation for Kazakh sentiment
Russian Sentiment Dataset	Russian	Training and evaluation for Russian sentiment
2GIS Reviews	Kazakh/Russian/Mixed	Final Kazakhstan-specific application

KazSAnDRA was used as the main Kazakh sentiment dataset, while the Kaggle Russian Sentiment Dataset was used for Russian-language examples.

Combined Multilingual Corpus

Source:

Reviews parsed from 2GIS for organizations located in Kazakhstan.

Review types:

- Cafes
- Clinics
- Shops
- Other local services

Main fields:

- text — raw review text
- rating or label — converted to binary sentiment
- Optional metadata: organization, city, category, timestamp

This dataset demonstrates the model on real Kazakhstan-related reviews.

Combined Multilingual Corpus

Preprocessing before merging:

- Cleaned column names
- Aligned label format
- Added explicit language labels

Language labels:

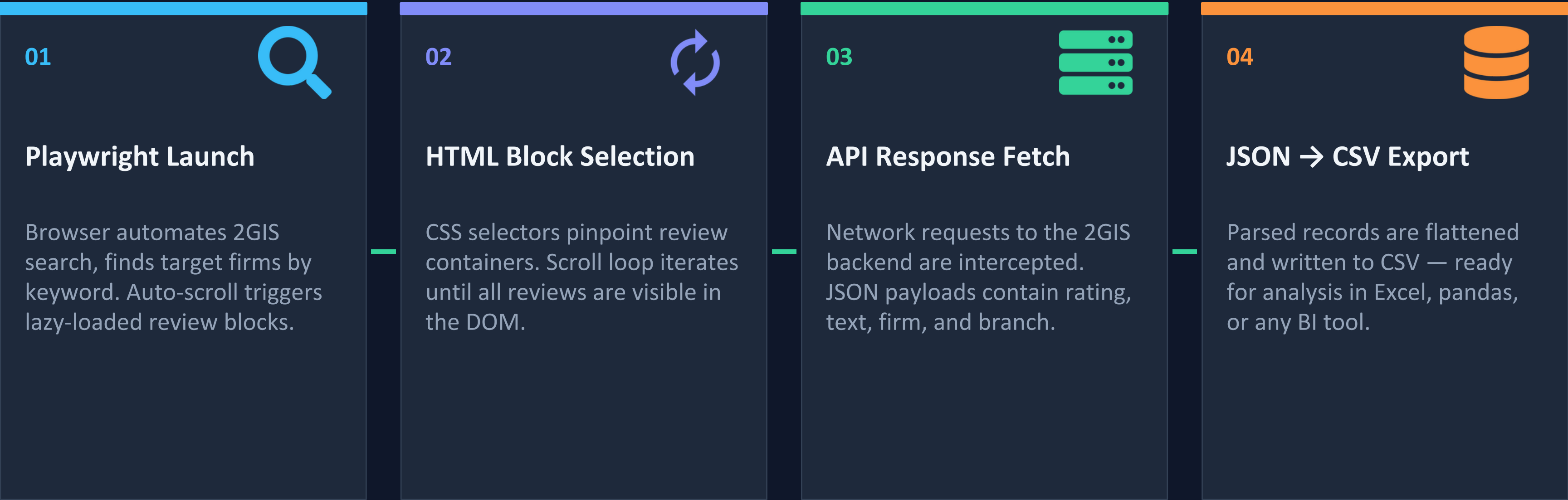
- kk — Kazakh
- ru — Russian

Why this is important:

Language labels allow the system to train separate language-specific models and support language-based routing during inference.

How It Works

Automated pipeline: browser automation → API interception → structured export



Key Features & Output

What makes this parser fast, reliable, and useful



Enhanced Performance

Headless Chromium with optimized selectors. Minimal wait times using smart DOM polling.



Automatic Firm Discovery

Keyword-driven search finds all matching branches automatically — no manual URL setup.



API-Level Data Capture

Intercepts XHR responses directly from 2GIS servers, bypassing fragile HTML scraping.



Structured CSV Output

Fields: rating, review text, firm name, branch. Drop-in compatible with Excel and pandas.

output.csv

firm

Магнум, Алматы

branch

ул. Достык 12

rating

4.5

text

Отличный сервис...

DATA CHALLENGES & CLASS IMBALANCE PROBLEM

Main challenges:

- Multilingual text
- Kazakh and Russian Cyrillic scripts
- Code-switching between languages
- Class imbalance
- Language imbalance
- Short and informal review texts

Why it is difficult:

Standard sentiment models often work best on one language, while this project deals with mixed multilingual real-world data.

Observed issue:

The datasets contain many more positive reviews than negative reviews.

Additional issue:

The combined corpus has more Russian examples than Kazakh examples.

Solution:

Undersampling was applied in two stages:

- Reduce Russian examples to avoid domination over Kazakh data
- Reduce positive Kazakh reviews to balance positive/negative labels

These steps were applied only to training splits, while validation and test splits preserved real-world distribution.

EXPLORATORY DATA ANALYSIS & DATA PREPROCESSING

EDA focused on:

1. Text length distribution
2. Sentiment label distribution

Key findings:

- Most reviews are short
- Most texts are under 200 characters
- A few reviews are much longer
- Positive reviews dominate the dataset

Visuals to insert:

- Histogram: Text Length Distribution
- Bar chart: Class Balance

The short length of most reviews supports using transformer sequence lengths of 128–256 tokens.

Steps applied:

- Removed HTML artifacts
- Removed extra whitespace
- Removed non-informative characters
- Lowercased text where appropriate
- Stripped leading/trailing spaces
- Avoided aggressive stemming or lemmatization

Reason:

Transformer tokenizers work better when the text is not over-normalized.

TRAIN/VALIDATION SPLIT & FEATURE ENGINEERING

Split strategy:

- Stratified train/validation split, for example 85/15.

Why stratified split was used:

- To preserve label balance across training and validation data.

Purpose:

- Ensure reliable evaluation while keeping enough training data for model learning.

For baseline models:

- TF-IDF features were used.

Feature types:

- Word-level n-grams
- Character-level n-grams

For transformer models:

- No manual feature engineering was used. Models processed cleaned raw text through pretrained tokenizers.

Main idea:

The project compares simple TF-IDF linear models with modern transformer models in a transparent setup.

BASELINE MODELS & TRANSFORMER MODELS

Classical models trained on the combined dataset:

1. Word TF-IDF + RidgeClassifier
2. Word TF-IDF + SGDClassifier
3. Word + Character TF-IDF + RidgeClassifier
4. Word + Character TF-IDF + SGDClassifier
5. fastText supervised model

Why these baselines were chosen:

They are lightweight, interpretable, and useful for comparison against transformer models.

Advanced models:

Model	Training Data	Purpose
mBERT	Kazakh subset	Kazakh-dominant reviews
ruBERT-tiny2	Russian subset	Russian reviews
mBERT multilingual	Full combined corpus	Mixed or uncertain texts

Training method:

Hugging Face Trainer API with early stopping
based on validation F1 macro.

LANGUAGE ROUTING STRATEGY

Main idea:

Instead of using one model for all reviews, the system first detects the language and then sends the text to the most suitable model.

Routing logic:

- Kazakh-dominant text → Kazakh mBERT
- Russian text → ruBERT-tiny2
- Mixed or uncertain text → multilingual mBERT

Benefit:

This improves handling of multilingual and code-switched reviews

EVALUATION METRICS

The models were evaluated using:

Metric	Meaning
Accuracy	Overall correct predictions
F1 macro	Average F1 across both classes
F1 weighted	F1 adjusted by class frequency
Precision	Correctness of positive predictions
Recall	Ability to find actual positives
ROC-AUC	Ranking quality between positive and negative classes

Primary metric:

F1 macro, because it treats positive and negative classes equally despite imbalance.

ERROR ANALYSIS & PRACTICAL APPLICATION

What to analyze:

- Incorrectly classified negative reviews
- Short reviews with little context
- Sarcasm or irony
- Mixed Kazakh-Russian texts
- Reviews with rating-text mismatch
- Informal spelling and slang

Purpose:

Show model limitations and explain where future improvements are needed.

How businesses can use the system:

- Monitor customer satisfaction automatically
- Identify negative review spikes
- Compare sentiment across branches
- Track sentiment by city or category
- Prioritize service quality improvements

Example use case:

A chain of cafes can automatically detect which branches receive more negative feedback and investigate the reasons.

Results

Model comparison

Model 	Accuracy	F1 Macro	ROC-AUC	Notes
Word TF-IDF + Ridge	0.8431	0.8431	—	Baseline
Word TF-IDF + SGD	0.8429	0.8429	—	Baseline
Word + Char TF-IDF + Ridge	0.8472	0.8472	—	Best baseline
Word + Char TF-IDF + SGD	0.8459	0.8459	—	Baseline
fastText supervised	0.8256	0.8256	—	Baseline
mBERT (kk only)	0.8364	0.8364	0.9042 	Transformer
ruBERT-tiny2 (ru only)	0.8787	0.8787	0.9504	Transformer
mBERT (all languages)	0.8531	0.8531	0.9275	Transformer

LIMITATIONS

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Kazakh data is smaller than Russian data

●

Class imbalance remains a challenge

●

Code-switching is difficult

●

Sarcasm is hard to detect

●

Some labels from star ratings may be noisy

Models may not generalize perfectly to all business categories

FUTURE IMPROVEMENTS



*Collect more
Kazakh reviews*



*Add neutral
sentiment class*



*Use better
language
detection*



*Try larger
multilingual
models*



*Add
explainability
tools*

*Analyze
sentiment by city,
category, and
organization*

*Build a
dashboard for
businesses*

CONCLUSION

Main conclusion *in operation*

The project demonstrates a multilingual sentiment analysis pipeline for Kazakhstan service reviews.

Key contribution *subscribers*

A language-routed approach that combines Kazakh-specific, Russian-specific, and multilingual transformer models.

Final value *raised*

The system can help businesses and analysts monitor customer satisfaction in a multilingual Kazakhstan environment.

Thank you!

QUESTIONS?